

Review

Persistent nociceptor hyperactivity as a painful evolutionary adaptation

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Chronic pain caused by injury or disease of the nervous system (neuropathic pain) has been linked to persistent electrical hyperactivity of the sensory neurons (nociceptors) specialized to detect damaging stimuli and/or inflammation. This pain and hyperactivity are considered maladaptive because both can persist long after injured tissues have healed and inflammation has resolved. While the assumption of maladaptiveness is appropriate in many diseases, accumulating evidence from diverse species, including humans, challenges the assumption that neuropathic pain and persistent nociceptor hyperactivity are always maladaptive. We review studies indicating that persistent nociceptor hyperactivity has undergone evolutionary selection in widespread, albeit selected, animal groups as a physiological response that can increase survival long after bodily injury, using both highly conserved and divergent underlying mechanisms.

Chronic neuropathic pain and associated nociceptor hyperactivity are nearly always assumed to be maladaptive

In evolutionary medicine, it is recognized that some disease symptoms commonly assumed to be maladaptive, such as diarrhea and fever, are in fact evolutionary adaptations promoting survival and reproductive success – notwithstanding the distress they cause [1]. One malady almost universally considered maladaptive is chronic **neuropathic pain** (see [Glossary](#)) [2,3], which contrasts with acute pain that has clear protective functions. In humans and rodents, many examples of neuropathic pain have been linked to persistent electrical hyperactivity in primary nociceptors (Figure 1 and Box 1) (reviewed in [2,4,5]). Because nociceptor activation evokes conscious pain in humans [6,7], and because nociceptor activity is sufficient and sometimes necessary for behaviors in rodents that indicate pain [8–11], continuing hyperactivity in nociceptors is recognized as a major source of persistent mammalian pain. Importantly, nociceptor hyperactivity is often generated ‘spontaneously’ at ‘ectopic’ sites near axonal injury or in nociceptor cell bodies rather than in the peripheral terminals where activation normally occurs (Figure 1). Moreover, both nociceptor activity and the associated pain sometimes persist after healing of an initiating injury. These properties are often taken as evidence of maladaptive pathology. Here we present evidence from diverse taxa indicating that persistent nociceptor hyperactivity can be an evolutionarily adaptive trait. While our focus is on nociceptor hyperactivity, alterations of various cell types in the CNS (central sensitization) are also important for persistent pain-like behavior, at least in mammals [2,12] and probably in insects (see below).

Persistent nociceptor hyperactivity can have many sources

In mammals, scores of extrinsic injury- and inflammation-related signals are known (Figure 2A). Many can directly excite and/or sensitize nociceptors, and some of these signals may be persistently released in chronic pain conditions [13–15]. Extrinsic signals detected by nociceptors may be released by dying cells [damage-associated molecular patterns (DAMPs)], released from (or detected by direct contact with) microbes [pathogen-associated molecular patterns (PAMPs)], and

Highlights

Injury to the nervous system can generate chronic pain driven by persistent hyperactivity of nociceptors, sensory neurons specialized to detect damaging stimuli.

Recent evidence argues against the near-universal assumption that persistent nociceptor hyperactivity and resulting chronic pain are always maladaptive.

Widespread conservation of mechanisms of nociceptor function and hyperactivity are well documented, as well as distinct adaptations to divergent nociceptive needs.

Persistent nociceptor hyperactivity could have evolved to drive painful hypervigilance during persistent physical impairment, enhancing survival by decreasing subsequent risk of predatory or aggressive attack.

Differences between rodent and human nociceptors are consistent with the possibility that, during relatively recent evolution, nociceptors increased their susceptibility to persistent hyperactivity, increasing the propensity of humans to chronic pain.

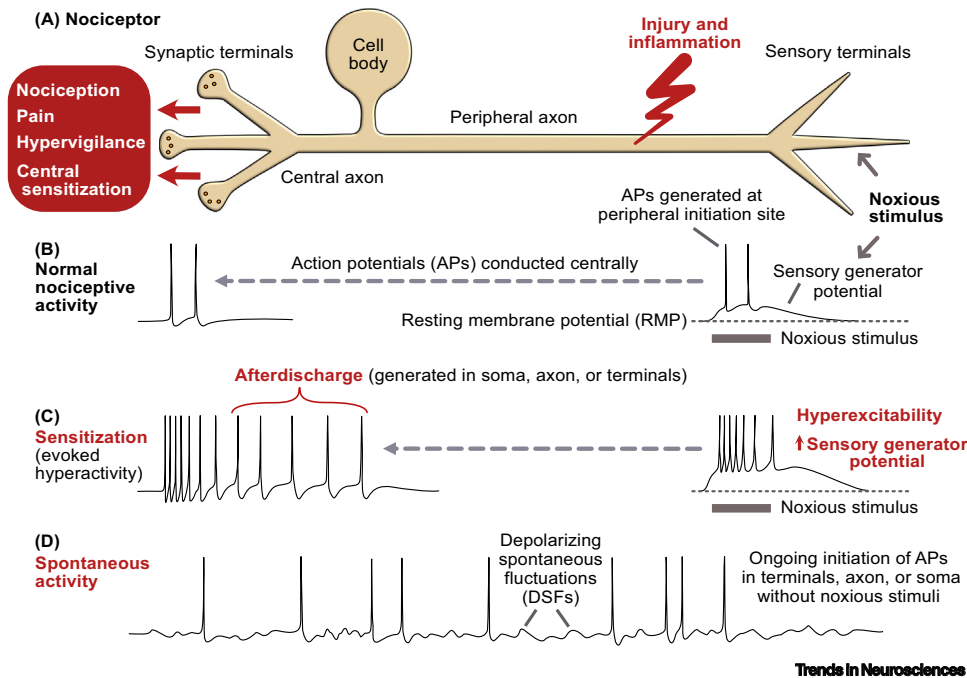
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Figure 1. Nociceptor hyperactivity. (A) Schematic of a primary nociceptor in vertebrates, gastropod molluscs, and leeches. The cell body is centrally located and distant from peripheral terminals where survivable injury and inflammation are most likely. In vertebrates, the cell body is in a ganglion near the CNS. In molluscs and leeches, nociceptor cell bodies are in ganglia within the CNS. (B) Normal nociceptive activity initiated by noxious stimulation of peripheral terminals produces a depolarizing sensory generator potential that may reach threshold for action potentials (APs). These APs are conducted to the CNS, sometimes evoking pain. (C) Illustration of sensitization of nociceptive activity evoked by the same noxious stimulus as in panel B, with the evoked hyperactivity following injury caused by a larger sensory generator potential, increased terminal excitability, and afterdischarge triggered by the evoked APs, which sometimes elicit pain, hypervigilance, and central sensitization. Experimentally, injection of pulses or ramps of depolarizing current (usually into the cell body) is often used to reveal hyperexcitability (Box 1). (D) Spontaneous activity after injury, defined as ongoing activity generated in the absence of concurrent extrinsic stimulation of a neuron [the spontaneous activity is sometimes generated by depolarizing spontaneous fluctuations (DSFs)] [100]. The term spontaneous activity is often used more loosely in referring to any nociceptor activity in the absence of evident ongoing noxious stimulation, including cases where intrinsic (cell autonomous) hyperexcitability and unobserved inflammatory signals combine to drive ongoing hyperactivity. Spontaneous activity and/or afterdischarge have been found in peripheral terminals, injured axons, and cell bodies of mammals and molluscs.

released by numerous cell types, including cells in the PNS and CNS and the innate and adaptive immune systems (releasing amino acids, biogenic amines, purines, eicosanoids, cytokines, growth factors, and other signals) [16–25]. Furthermore, various cell types, including immune cells, release signals such as endogenous opioids that inhibit nociceptor activity [26]. In principle, continuing disinhibition by reduced release of inhibitory signals could promote persistent hyperactivity (Figure 2A). In invertebrates, far less is known about the extrinsic signals that promote persistent nociceptor hyperactivity and the cells releasing them [27], although sensitizing signals coming from immune cells have been implicated in the marine snail *Aplysia californica* [28,29]. A fascinating question that exceeds the scope of our review concerns the evolution of immune systems and inflammatory mediators, for example [30–33]. Thus far, little is known about the important likelihood of coevolution of nociceptors and immune cells [34].

Apart from extrinsic signals detected by nociceptors, intrinsic, cell-autonomous sources of persistent hyperactivity after bodily injury are also prominent in mammalian nociceptors (Figure 2B) and in

Box 1. Hyperactivity and hyperexcitability are not the same

The terms hyperexcitability and hyperactivity are often used interchangeably, but hyperactivity is arguably more important physiologically and evolutionarily. In studies of neuronal plasticity involving alterations of AP function, **the most common experimental indicator of an altered electrophysiological state is modified electrical excitability – either hyperexcitability or hypoexcitability**. Hyperexcitability is demonstrated when electrical stimulation more readily triggers an AP. Experimental delivery of depolarizing current, either in a series of increasingly depolarizing current steps or in response to a smooth depolarizing ramp of current, yields two indicators of hyperexcitability: generation of an AP by stimulation with less stimulating current (sometimes referred to as a decrease in rheobase) and/or AP generation at a more negative threshold potential. Under natural conditions, hyperexcitability would make it more likely that a physiological stimulus such as a sensory generator potential (see Figure 1 in main text) or synaptic potential would reach AP threshold. For most neurons, the frequency and pattern of APs appear to be the critical physiological endpoints of excitability because they determine a neuron's immediate effect on its postsynaptic targets and can control activity-dependent synaptic plasticity.

Hyperexcitability usually promotes hyperactivity, but many other intrinsic and extrinsic states of a neuron and its inputs will determine whether a neuron is hyperactive and what its level and pattern of activity are. As shown with examples in Figure 2 in main text, numerous drivers may contribute to hyperactivity in mammalian nociceptors after bodily injury, including intrinsic hyperexcitability, intrinsic hypersensitivity to extrinsic excitatory chemical signals, increased exposure to extrinsic excitatory signals, decreased exposure to extrinsic inhibitory signals, and hyposensitivity to inhibitory signals.

Evolutionary selection could change any or all of the intrinsic and extrinsic drivers of persistent nociceptor hyperactivity to make some species more or less likely to experience persistent pain-related states driven by continuing activity of nociceptors. The accessibility of nociceptors in diverse animal taxa enables direct inquiry into potentially adaptive functions of persistent nociceptor hyperactivity, as well as into similarities and differences in the underlying mechanisms.

some invertebrate nociceptors. Intrinsic alterations in identified nociceptors are readily compared across taxa, a major aim of this review. **Intrinsic alterations include biophysical changes producing hyperexcitability**, as well as alterations making a nociceptor either more sensitive to excitatory or sensitizing extrinsic signals, less sensitive to inhibitory signals, or more sensitive to normal body temperature. Importantly, intrinsic and extrinsic signaling often work cooperatively. **Indeed, because nociceptors are normally electrically silent and not hyperexcitable, hyperactivity elicited by injury outside the nociceptor necessarily involves extrinsic triggers**. These can have complex, long-lasting effects. For example, transient exposure to serotonin in *Aplysia* or rats triggers an intrinsic hyperactive state that lasts minutes or hours after the signal is removed, but prolonged or repeated exposure to serotonin or other inflammatory signals may sustain hyperactivity for longer periods, or induce an enduring, memory-like hyperactive state [35–39].

Primary nociceptors and evolution

Because nociceptors detect injury, and injury affecting survival and reproduction can occur in any animal, cells with this function should be widespread. **Indeed, nociceptors have been identified in diverse animals, including roundworms, fruit flies, leeches, snails, squid, fish, birds, rodents, cats, pigs, monkeys, and humans [40,41]**. Nociceptors thus provide an opportunity for broad comparative studies into evolutionary selection pressures for nociceptive plasticity and the roles of shared and divergent mechanisms. Adding to their evolutionary interest, some adaptive cellular responses to injury now found in nociceptors may have antedated the earliest animals, **possibly shaping core mechanisms later used for basic neural function as well as nociception [42–44]**. Nociceptors probably evolved very early, but may have also evolved independently in different lineages in response to common injury-related selection pressures.

Taxa in which persistent nociceptor hyperactivity may not be functionally important

Insight into the adaptive roles of a physiological trait can be gained by comparing the animals and ecological conditions in which the trait is prominent with those in which it appears to be weak or absent. We begin with nociceptors exhibiting little evidence of persistent hyperactivity after bodily injury.

Glossary

Afterdischarge: APs generated in response to a preceding AP or burst of APs, prolonging the electrical activity.

Depolarizing spontaneous fluctuation (DSF): brief, intrinsically generated depolarizing fluctuations of RMP. In nociceptors, some DSFs may intermittently reach AP threshold, contributing to spontaneous activity under hyperactive conditions.

Hyperactivity: state of increased discharge of APs resulting from drivers that are intrinsic and/or extrinsic to a neuron.

Hyperexcitability: electrophysiological state in which the likelihood of generating at least one AP by depolarizing stimulation is increased. **Hyperexcitability is electrical hypersensitivity**, and represents one type of sensitization.

Hypersensitivity: state of increased responsiveness to a biological signal (sensory, chemical, or electrical). The term is almost synonymous with sensitization, although in the pain field it tends to be used more descriptively and less mechanistically than sensitization.

Neuropathic pain: pain produced by injury or disease of any part of the nervous system.

Nociceptor: a primary sensory neuron that is selectively activated by stimuli causing actual or incipient tissue injury, or by signals reliably associated with tissue injury (e.g., inflammatory signals).

Resting membrane potential (RMP): electrical potential difference across the plasma membrane in the absence of APs, sensory potentials, or synaptic potentials. If DSFs occur, they are considered transient components of RMP.

Sensitization (see also hypersensitivity): a state in which a system (typically a neuron, neuronal pathway, or animal) exhibits increased responsiveness to biological signals.

Spontaneous activity: in electrophysiology, the ongoing generation of APs by a cell without concurrent extrinsic sources of depolarization. Because it is often impractical to identify background excitatory or sensitizing contributions to ongoing AP discharge, ongoing discharge generated in neurons in the absence of observed sensory generator potentials or synaptic potentials is commonly referred to as spontaneous activity.

(A) Extrinsic sources of nociceptor hyperactivity (mammalian examples)

↑ Excitatory or sensitizing signals (↑ exposure to):

Damage-associated molecular patterns (DAMPs)
(e.g., K^+ , H^+ , ATP, glutamate, heat shock proteins)

Pathogen-associated molecular patterns (PAMPs)
(e.g., LPS, hemolysin, double-stranded RNAs)

Cytokines
(e.g., TNF- α , IL-1 β , CXCL1, CCL2, MIF)

Biogenic amines (e.g., serotonin, histamine, NE)

Eicosanoids (e.g., prostaglandins, leukotrienes)

Peptides (e.g., bradykinin)

Growth factors (e.g., NGF, GDNF)

↓ Inhibitory signals (disinhibition, ↓ exposure to):

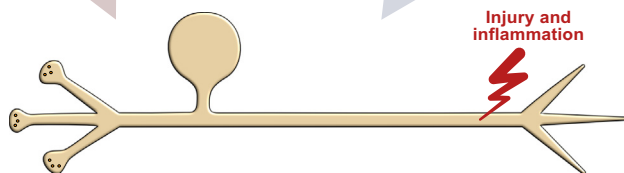
Inhibitory cytokines (e.g., IL-10)

Endogenous opioids
(e.g., enkephalins, endorphins, dynorphins)

Other inhibitory peptides (e.g., somatostatin)

Endocannabinoids

GABA



(B) Intrinsic alterations promoting nociceptor hyperactivity

Intrinsic hyperexcitability

↑ Sensory generator potentials
(e.g., more TRP channels)

↑ Membrane resistance
(e.g., fewer open K^+ channels)

↓ RMP (depolarized, e.g., fewer open K^+ channels)

↓ AP threshold (e.g., hyperpolarized VGSC activation)

↑ AP discharge (e.g., more VGSCs, fewer VGPCs)

↑ DSF amplitude and frequency
(e.g., more Na^+ and Ca^{2+} channels open at RMP, fewer open K^+ channels)

Intrinsic sensitivity alterations

↑ Sensitivity to excitatory or sensitizing chemical signals
(e.g., more stimulatory GPCRs, enhanced downstream signaling)

↑ Sensitivity to normal temperatures
(e.g., altered thermosensitive TRP channel expression)

↓ Sensitivity to inhibitory chemical signals
(disinhibition, e.g., fewer inhibitory GPCRs, suppressed downstream signaling)

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Figure 2. Sources of nociceptor hyperactivity following injury. (A) Left: Some of the many extrinsic chemical signals implicated in mammals to excite and/or sensitize nociceptors. Many of these have multiple sources (e.g., glutamate and ATP are DAMPs released from ruptured cells but also are secreted exocytotically from various cell types including neurons). Right: Hyperactivity may also be promoted by reducing exposure to inhibitory chemical signals (disinhibition). (B) In principle, nociceptor hyperactivity can be promoted by intrinsic hyperexcitability via any or all the listed electrophysiological alterations [100,104]. Hyperactivity may also be promoted by increasing a nociceptor's intrinsic sensitivity to excitatory or sensitizing sensory stimuli (including normal body temperature) or chemical signals, and by decreasing intrinsic sensitivity to inhibitory chemical signals. Abbreviations: AP, action potential; CCL2, chemokine (C-C motif) ligand 2; CXCL1, chemokine (C-X-C motif) ligand 1; DSF, depolarizing spontaneous fluctuation; GDNF, glial-derived neurotrophic factor; GPCR, G protein-coupled receptor; IL-1 β , interleukin-1 β ; IL-10, interleukin 10; LPS, lipopolysaccharide; MIF, macrophage migration inhibitory factor; NE, norepinephrine; NGF, nerve growth factor; RMP, resting membrane potential; TNF α , tumor necrosis factor α ; TRP, transient receptor potential; VGPC, voltage-gated potassium channel; VGSC, voltage-gated sodium channel.

Nematodes

Caenorhabditis elegans has polymodal nociceptors that repeatedly respond to noxious mechanical or heat stimuli and receive neuromodulatory input to prevent habituation to noxious stimuli [45]. Unusually, *C. elegans* neurons primarily signal by graded depolarizing responses rather than action potentials (APs) (Figure 1) [46]. In principle, persistent depolarization without APs

might maintain nociceptive sensitization, but neither persistent depolarization nor acute or persistent AP generation has been reported in nematode nociceptors after injury. While a lack of observed nociceptor hyperactivity in this group (or any other taxon) does not prove it is absent, the short lifespan of *C. elegans* (about 20 days) and its tiny body size might be factors that make persistent nociceptor alterations relatively inconsequential (Table 1), despite the worms being subject to predation [47].

Insects

Larvae of the fruit fly *Drosophila melanogaster* show behavioral sensitization after tissue injury, including a lowering of threshold for heat- or mechanically induced defensive rolling behavior following UV irradiation of the epidermis [48,49]. The behavioral sensitization was linked to various molecular changes in heat-sensitive nociceptors [50], and accompanied by an increase in the range of temperatures that evoked continuing generation of APs [51]. In rodents, 'spontaneous' pain may be driven by continuing activity in heat-sensitive nociceptors resulting from a lowering of threshold into the range of normal body temperature after injury [52]. However, for heat allodynia in *Drosophila*, UV-induced **hypersensitivity** of defensive behavior was recently found to depend on the acquisition of heat sensitivity by central neurons rather than enhanced heat sensitivity in nociceptors [53]. Thus, if injury-induced nociceptor hyperactivity occurs (which has not been demonstrated physiologically), it appears to contribute little to the UV-induced behavioral sensitization. Results from adult *Drosophila* also point to alterations in the CNS underlying injury-induced hypersensitivity of a defensive behavior, without obvious contributions from injured primary nociceptors [54]. Amputation of one leg lowered the threshold for jumping escape to heat stimulation for at least 3 weeks. Initial nociceptor activation was necessary for inducing permanent central alterations (excitotoxic loss of GABAergic inhibitory neurons, producing central sensitization) and behavioral

Table 1. Summary of evidence for functionally important, persistent nociceptor hyperactivity and complementary neural and immune alterations after injury in different taxa

Phylum (genus or species)	Injury-induced nociceptor hyperactivity?	Survival link?	Required cell signaling in nociceptors	Complementary alterations in other cell types	Refs
Nematoda (<i>C. elegans</i>)	Not observed	No	Probably none	Unknown	[45,46]
Arthropoda (<i>Drosophila</i> – fly)	Unclear	Unknown	Unclear	Sensitization of central interneurons, central disinhibition	[50–54]
Mollusca (<i>Aplysia</i> – snail)	Hyperexcitability, afterdischarge	Unknown but plausible	cAMP, PKA, CREB, ERK, MNK, eIF4E	Sensitization of central interneurons, central disinhibition	[35,55,66–74,106,109]
Mollusca (<i>Doryteuthis</i> – squid)	Hyperexcitability, afterdischarge, spontaneous activity	Direct evidence	Unknown	Unknown	[75,86,87]
Chordata (<i>Mus</i> or <i>Rattus</i> – laboratory mice or rats)	Hyperexcitability, afterdischarge, spontaneous activity	Indirect evidence	cAMP, PKA, CREB, ERK, MNK, eIF4E	Sensitization of central interneurons, central disinhibition, glial and immune cell activation	[2–5,11,12,17,25,26,52,77,78,81,82,88,93,100–103,107,108,110–112,114–118]
Chordata (<i>Heterocephalus</i> – naked mole-rat)	Not observed	No	Probably none	Unknown	[59,60,63–65]
Chordata (<i>Homo sapiens</i>)	Hyperexcitability, spontaneous activity	Unknown but plausible	Unknown	Probable sensitization of central interneurons, central disinhibition, glial activation, direct evidence for immune cell activation in DRG	[2,7,13,24,79,80,83,84,104]

sensitization to heat test stimuli. However, no nociceptor alterations were described other than loss of all nociceptors in the amputated limb [54]. The use of amputation highlights an unusual feature of *Drosophila* nociceptors that may be important for the evolution of injury-induced persistent nociceptor hyperactivity. Unlike in leeches, *Aplysia*, squid, and vertebrates, where nociceptor cell bodies occur within centralized ganglia, *Drosophila*'s nociceptor cell bodies are located just beneath the epidermis where they can easily be killed by superficial tissue injuries. This anatomical feature in insects should prevent persistent site-specific nociceptor sensitization after injury [55]. It is consistent with the lack of enhanced sensory activity recorded from nerves innervating a previously injured region in larvae of the moth *Manduca sexta* [56], and with central mechanisms implicated in generalized sensitization and hypervigilance in *Drosophila* [53,54].

Mole-rats

Most investigations of persistent nociceptor hyperactivity in mammals have used laboratory rats (specifically, *Rattus norvegicus*), and mice (*Mus musculus*), which show complex alterations of both behavior and nociceptor function after injury. Quite different nociceptive reactions have been found in naked mole-rats (*Heterocephalus glaber*), which show distinctive traits related to their subterranean habitat [57]. Three differences have been found between nociceptors of naked mole-rats and laboratory rodents that are consistent with limited injury-induced nociceptor hyperactivity in mole-rats.

First, mole-rats have far fewer cutaneous nociceptors. In most mammals, the skin is innervated by approximately four times as many unmyelinated C-fibers as myelinated A-fibers, and most cutaneous C-fibers are nociceptors [58]. In adult naked mole-rats, the ratio of cutaneous C- to A-fibers is ~1:1 [59,60]. Fewer nociceptors in a subterranean animal might represent an evolutionary response to reduce the energy costs of maintaining defensive neurons that are less useful in an environment where predation is reduced, as has been suggested for a central defensive neuron, the Mauthner cell, in a cave fish species [61,62].

Second, nociceptors in naked mole-rats exhibit adaptations that limit their ability to drive pain under inflammatory conditions. Presumed nociceptors in mammals are often identified by sensitivity to capsaicin, the pain-evoking ingredient of chili peppers, and by expression of the capsaicin receptor, transient receptor potential vanilloid 1 (TRPV1). In other mammals, TRPV1+ nociceptors are important for driving inflammatory pain. Unexpectedly, in naked mole-rats, the synapses of TRPV1+ sensory neurons are distributed randomly throughout the dorsal spinal cord [59], rather than being concentrated in the superficial dorsal horn of the spinal cord, which is critical for relaying nociception in other mammals. Another adaptation reduces acid sensitivity. Tissue injury and inflammation are accompanied by tissue acidosis, and most mammals utilize a plethora of acid-sensitive receptors and ion channels to activate nociceptors and drive inflammatory pain [63]. However, naked mole-rats live in large colonies in poorly ventilated burrows, and thus experience high CO₂ levels. CO₂ can produce carbonic acid, releasing protons that activate acid receptors in naked mole-rats and other rodents. Surprisingly, acid fails to evoke APs or consequent pain behavior in naked mole-rats. This is because a mutation in voltage-gated Na⁺ channel Na_v1.7 enables protons to block the channel and prevent AP generation under acidic conditions [64].

A third adaptation is reduced function of the naked mole-rat tropomyosin receptor kinase A (TrkA) for nerve growth factor (NGF), which is released in mammals during injury and inflammation. This TrkA hypofunctionality results in an absence of NGF-induced hyperalgesia in naked mole-rats and might also contribute developmentally to reduced nociceptor number. These adaptations in the naked mole-rat and additional adaptations in related mole-rats [65] should reduce the

likelihood of persistent nociceptor hyperactivity being induced by injury and inflammation (although this prediction has yet to be tested directly). Loss of nociceptor hyperactivity might have been promoted during evolution by the relative safety from predation afforded by subterranean environments.

Taxa in which persistent nociceptor hyperactivity is likely to be functionally important

All the species in which persistent nociceptor hyperactivity has been observed after nerve or tissue injury have probably been subject to high levels of predation in their recent evolutionary history. In each case, persistent hyperactivity is generated near a site of injury and sometimes also in nociceptor cell bodies distant from the body surface (Figure 1 and Table 1).

Gastropod molluscs

A. californica lacks a shell and is one of the larger invertebrates, reaching weights of ~1 kg in its 2-year lifespan. Identified *Aplysia* nociceptors are activated preferentially by pinching or crushing stimuli and are unresponsive to chemical or thermal stimuli. Hyperactivity lasting days to months after experimental injury (nerve crush or wounding that transects major nerve branches) is expressed as reduced AP threshold and **afterdischarge** (Figure 1C) generated in nociceptor peripheral terminals [66], the injured axon [67], and the cell body [68]. The function of nociceptor cell body hyperactivity in *Aplysia* is, via afterdischarge, to amplify the number of APs reaching central synaptic terminals, thereby enhancing defensive behavioral responses [69]. Many invertebrates, including *Aplysia* and leeches, lack image-forming eyes and depend upon chemical and mechanical senses to detect predatory threats. In *Aplysia*, nociceptive sensitization after injury reduces nociceptor mechanical threshold into the innocuous range [70], which may promote escape at the first contact with slowly moving predatory gastropods [71]. *Aplysia* also exhibit short-term central sensitization expressed in identified interneurons [72,73], and *in vitro* evidence is consistent with long-term central sensitization that may increase serotonin release onto nociceptors [74].

Cephalopod molluscs

Nociceptors that respond to intense mechanical stimuli and fail to respond to heat have also been identified in the squid *Doryteuthis pealeii*. Activity recorded in fin-innervating nerves showed that nociceptors are activated by forces that can damage the fin, and, following noxious activation, the nociceptor mechanical activation threshold decreased dramatically (~50-fold) [75]. Two findings in this study were unexpected. First, injury produced either experimentally to the fin *in vivo* or more naturally to various body regions (caused by attacks from other squid) produced **spontaneous activity** (Figure 1D) lasting at least 24 h after injury. This remains the only reported invertebrate example (to our knowledge) of injury-induced spontaneous activity in primary afferent neurons. Second, after injury to one side of the animal, both nociceptor sensitization to mechanical stimulation and spontaneous activity were found in nociceptors innervating the contralateral fin. This generalized hyperactivity contrasts with site-specific hyperactivity in mammalian and gastropod nociceptors, which is usually restricted to neurons with innervation close to injured regions.

Mammals

Most studies of persistent nociceptor hyperactivity in mammals have utilized rat or mouse neuropathic pain models (reviewed by [2,4,5]). Hyperactivity can manifest in nociceptors as increased generation of APs evoked by extrinsic signals or generated by intrinsic alterations (spontaneous activity) (Figures 1C,D, 2A,B, and Box 1). Cell body hyperactivity is typically assumed to be an uncommon and less important parallel to hyperactivity generated in peripheral terminals. The more severe an injury, however, the more likely peripheral nerve branches are to be crushed or

cut, disconnecting affected nociceptors from their normal sources of excitation. Peripherally disconnected nociceptors may still contribute ongoing information about injury because sensory receptors and ion channels accumulate at the tips of cut axons, which along with other alterations, can impart sensitivity, hyperexcitability, and spontaneous activity to axons at the injury site [76–78]. Ongoing pain in humans from peripheral injury or neuropathy [79] can be eliminated rapidly by blocking peripheral nerve conduction with a non-selective voltage-gated sodium channel (Na_v) inhibitor, lidocaine, which also can partially ameliorate pain from amputation [80]. In addition, persistent spontaneous activity may be generated in the cell body [81,82]. For example, lidocaine delivered to human dorsal root ganglia (DRG) innervating an amputated limb at concentrations that block local AP generation, without blocking conduction through the ganglia, rapidly and reversibly relieved phantom limb pain [83] (Figure 3, Key figure, and Table 1), providing strong evidence for the clinical importance of spontaneous activity generated in sensory neuron cell bodies. Spontaneous activity also occurs in dissociated DRG neurons from cancer patients with neuropathic pain who have had DRGs removed during thoracic vertebrectomy [84].

An experimental advantage of cell-body generation of hyperactivity in mammalian (and gastropod) nociceptors is that the mechanisms can be investigated more directly than is possible in other neuronal compartments. Although dissociation produces highly abnormal test conditions, spontaneous activity induced *in vivo* is surprisingly resistant to dissociation of rodent DRG neurons. In a DRG compression model, similar spontaneous activity was recorded from a DRG neuron *in vivo* before and after complete isolation [85], and in a spinal cord injury model, remarkable parallels were found in the pattern and discharge frequency of spontaneous activity generated in cell bodies of probable nociceptors *in vivo* and after dissociation [82].

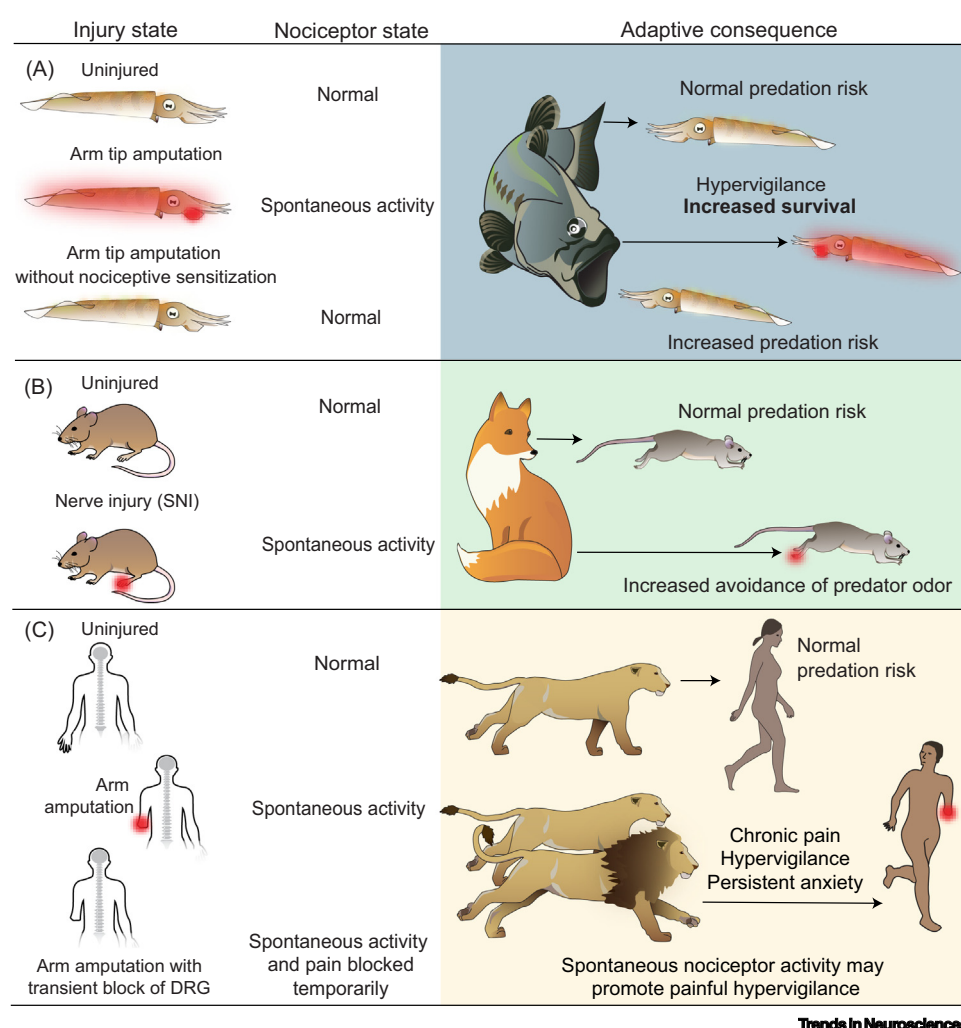
Evidence that pain-like behavior associated with nociceptor hyperactivity is adaptive in cephalopod molluscs and mice

As mentioned, injury in squid induces long-lasting nociceptor spontaneous activity, which was suggested to drive a state of generalized hypervigilance [75,86]. The adaptiveness of this state was tested by examining the effects on survival of squid in staged interactions with natural predators (Figure 3A) [87]. A relatively minor injury, distal amputation of one of a squid's eight arms, produced immediate defensive behavior and sensitization of responses to tactile and visual stimuli, but no lasting effects on motor function [86]. When exposed to predatory sea bass (*Centropristis striata*), squid with amputation injury were more likely to be attacked than uninjured squid, but they also began their escape behavior sooner than uninjured squid [87]. Briefly blocking nociceptor activation/sensitization during amputation by transient delivery of an anesthetic dose of MgCl_2 resulted in injured squid later failing to show early responses to approaching sea bass and being less likely to escape. This result demonstrates how nociceptive sensitization enhances evolutionary fitness by increasing survival during predatory encounters and, when combined with electrophysiological observations, it suggests that persistent nociceptor spontaneous activity can drive a hypervigilant state that protects animals from attack during periods of increased vulnerability (Table 1).

Analogous results [88] were found in mice experiencing chronic pain in a spared nerve injury model (Figure 3B). When neuropathic mice had to choose between long and short routes to a food reward, they were more likely than uninjured mice to avoid the short route if it exposed them to the smell of fox urine, suggesting that continuing awareness of their injury (perhaps driven by nociceptor spontaneous activity [25]) influenced their behavioral choices during elevated predatory risk [88]. Persistent injury-induced hypervigilance in animals appears functionally similar to anxiety in humans [89], and anxiety or anxiety-related behaviors are a common comorbidity of chronic pain in humans and rodents [90].

Key Figure

Persistent nociceptor hyperactivity can be adaptive



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Figure 3. (A) Squid survival is enhanced by nociceptor activity after injury. Peripheral tissue injury causes long-lasting nociceptor hyperactivity expressed over much of the body surface [75] (red). In staged 30-min encounters with fish predators [87], ~75% of uninjured subjects survived. Nearly half of squid previously receiving a minor arm amputation survived. Although selectively targeted by the fish, injured squid began their escape behavior farther from approaching fish (indicated by arrow length) than uninjured squid [87]. Squid that were injured without nociceptive sensitization and persistent nociceptor hyperactivity (prevented by transient block of nociceptor activity during amputation) waited longer to begin escape, with only ~20% surviving, indicating the survival benefit of persistent nociceptor hyperactivity after injury [86]. (B) Enhanced avoidance of a predator cue by mice with neuropathic pain. Mice with a spared nerve injury model exhibit nociceptor hyperactivity generated in the hindpaw (red) [138] and dorsal root ganglia (DRG) [25, 139] (not shown). When exposed to fox urine, injured mice chose a route that took them farther from the predator odor [88]. **This suggests that injury-induced nociceptor hyperactivity promotes hypervigilance and increased risk aversion.** (C) Speculative benefit of chronic nociceptor hyperactivity in ancestral humans at high risk for predation after injury (e.g., amputation injury in the illustration). The top two rows parallel the arguments for squid illustrated in panel A. The bottom row illustrates a situation of transient clinical blockade of DRG action potentials (APs) in amputees that blocks ongoing pain [83]. If spontaneous activity in human nociceptors persistently drives hypervigilance as well as pain (as suggested by anxiety being a comorbidity of pain [90]), this hyperactivity should be protective for injured individuals who are more likely to be targeted by predators.

Are the hypervigilant states driven by nociceptor hyperactivity in cephalopods associated with motivational properties resembling those associated with affective pain in mammals? This question has not yet been addressed experimentally in squid, to our knowledge, but a study in *Octopus bocki* [91] using the noxious stimulus acetic acid showed that nociceptive states with pain-like aversive features occur in at least one cephalopod species. Specifically, octopuses remembered a chamber in which an arm was injected with acetic acid, and subsequently avoided it (referred to as conditioned place aversion). When acetic acid injection was followed by lidocaine injection into the same site immediately prior to placement in a different chamber, octopuses later chose to spend time in the chamber associated with lidocaine's block of ongoing nociceptive activity (i.e., conditioned place preference). Thus, the octopus's voluntary behavior revealed a pain-like link between flexible cognitive processing (including complex associative learning, memory, and decision-making) and an aversive state induced by noxious stimulation. Although recordings could not be made from primary afferent neurons in the octopus arm, downstream electrical activity recorded in a central pathway from the injected arm to the brain continued for at least 30 min after acetic acid injection. The immediate block of this activity by lidocaine delivered to the acetic acid injection site is consistent with ongoing nociceptor activity generated at an injury site driving pain-like motivational and cognitive processes that enable conditioned place aversion and place preference in octopuses. These findings [91] parallel observations in rodents where spontaneous nociceptor activity is associated with a tonic aversive state [11,92].

Indirect evidence consistent with persistent nociceptor hyperactivity being adaptive in mammals

Several lines of evidence are consistent with the adaptiveness of spontaneous activity generated persistently in the cell bodies of mammalian nociceptors after injuries severe enough to increase an individual's vulnerability to attack [93]. First, mammalian primary sensory neuron cell bodies are likely to survive serious peripheral injury because they are protected by their central location far from their receptive fields (Figure 1A), and by the bone and dura covering sensory ganglia. Second, the cell bodies are uniquely placed to integrate many injury signals (Figure 2) that may induce or persistently maintain hyperactivity [93–95]. Such signals may be intrinsic to injured neurons and conveyed electrically or by axonal transport from sites of peripheral damage. In addition, numerous extrinsic signal molecules may be conveyed to nociceptor cell bodies in the cerebrospinal fluid and the blood (because of the absence of an effective vascular permeability barrier in the ganglia). Peripheral or central tissue injury may also lead to local release of extrinsic signals near nociceptor cell bodies (from satellite glial cells and resident or infiltrating immune cells [95–97]) and near sensory neuron presynaptic terminals within the spinal dorsal horn (from postsynaptic neurons and spinal glia [93,94]). While these anatomic and signaling features have many biological explanations, a functional consequence is that they enable nociceptor cell bodies to survive peripheral injury, to integrate disparate sources of information about injury severity in tissues that may have been disconnected from the CNS, and through persistent spontaneous activity to inform the CNS about continuing peripheral dysfunction.

A third line of evidence comes from the complex, complementary physiological mechanisms found to drive hyperactivity in the nociceptor cell body, providing some evidence of 'functional design' by natural selection [98,99]. Specifically, in terms of membrane potential, there are three possible alterations that can result in AP generation in the absence of excitation from fast synaptic potentials (which mammalian nociceptors do not receive) or sensory receptor potentials (Figure 1B, which are unlikely to be generated in the cell body): depolarization of **resting membrane potential (RMP)** to reach AP threshold, hyperpolarization of AP threshold to reach RMP, and enhancement of **depolarizing spontaneous fluctuations (DSFs)** of RMP (Figure 1D) to bridge the gap between RMP and AP threshold. Recordings from dissociated

rodent nociceptors in a variety of different persistent pain models showed that all three properties are altered in these neurons compared with control conditions to promote spontaneous activity [100–103]. Spontaneous activity and all three underlying alterations were also found in probable nociceptors isolated from human cancer patients suffering from neuropathic pain [84,104]. Importantly, these complementary alterations occurred almost exclusively in nociceptors taken from ganglia innervating dermatomes reported by the patients to have ongoing pain.

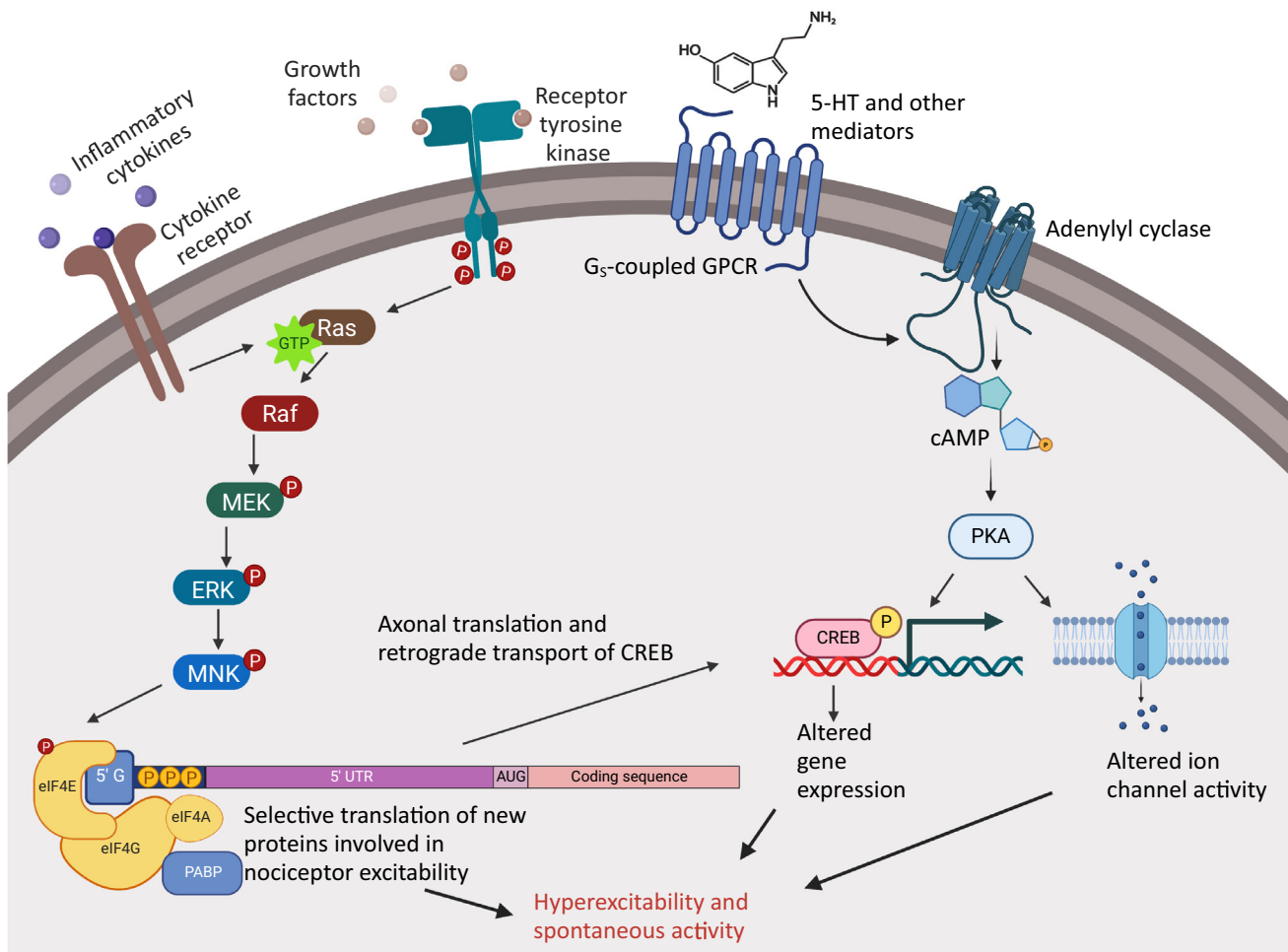
RMP, AP threshold, and DSFs in nociceptors each involve numerous ion channels, and the interactions between membrane potential, channel activity, and associated intracellular signaling are highly complex. Although spontaneous activity promoting neuropathic pain may sometimes be a purely pathological side effect of various insults to the body and nervous system, the intricate, functionally cohesive electrophysiological alterations that underpin spontaneous activity suggest that some damaging conditions may recruit a persistent, specialized state that was selected during evolution to ensure that nociceptor hyperactivity continues after injury severe enough to produce lasting physical impairment detectable by predators and competitors. Continuing pain would promote protective behavior (including hypervigilance, sometimes manifested in mammals as anxiety [90]) during increased vulnerability (Table 1), which could persist chronically for severe injuries such as amputation (Figure 3A,C).

Shared signaling mechanisms support nociceptor hyperactivity in gastropod molluscs and rodents

One of the interesting evolutionary aspects of persistent nociceptor hyperactivity is that it is found in two distantly related animal phyla, chordates and molluscs, suggesting common functions and potentially ancient mechanisms. However, even within placental mammals, nociceptor physiology shows clear divergences, raising questions about whether fundamental hyperactivity mechanisms are shared across the enormous phylogenetic distances between invertebrates and mammals.

Indeed, nociceptors in rodents and *Aplysia* share fundamental mechanisms important for synaptic plasticity [105] and hyperactivity. In *Aplysia*, long-lasting hyperactivity in dissociated nociceptors induced by serotonin was found to require protein synthesis [106]. Persistent axonal hyperactivity induced in *Aplysia* by nerve injury or nerve depolarization using elevated extracellular K⁺ to mimic the depolarization produced by axotomy [67], or serotonin application to a nerve [35], also requires protein synthesis, which is controlled by the mechanistic target of rapamycin pathway. Detailed studies in mice showed that activity-dependent mRNA translation regulated by mitogen-activated protein kinase-interacting kinase (MNK) phosphorylation of eukaryotic translation initiation factor 4E (eIF4E) is necessary for nociceptor hyperexcitability after peripheral nerve injury or inflammatory signals [107,108]. *Aplysia* has MNK1 and eIF4E proteins [109], with conservation of the eIF4E serine residue that is phosphorylated by MNK1 and required in mice for nociceptor hyperactivity induced by inflammatory mediators [107]. A selective MNK inhibitor was shown to potently inhibit the axonal hyperexcitability induced by depolarization of an *Aplysia* nerve, as do general inhibitors of tyrosine receptor kinases (upstream activators of MNKs) and of extracellular receptor kinases (ERKs, which activate MNKs) [109] (Figure 4).

A cell signal that plays prominent but complex roles in persistent hyperexcitability in *Aplysia* nociceptors is cAMP and its downstream signals protein kinase A (PKA) and the transcription factor, cAMP response element-binding protein (CREB) [110–112] (Figure 4). Cyclic AMP has long been known to increase mammalian nociceptor excitability [113] and cAMP signaling is essential for maintaining rat nociceptor hyperactivity months after spinal cord injury, via PKA [114], exchange protein activated by cAMP [101], and indirect positive feedback through



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Figure 4. Shared signaling pathways driving nociceptor hyperactivity in *Aplysia* and rodents. Two general cell signaling pathways have been implicated directly and indirectly in the maintenance and induction of persistent nociceptor hyperactivity both in a gastropod mollusc and in mammals: Ras-MNK signaling and cAMP-PKA signaling. These pathways are more complex than shown (e.g., cAMP has additional effectors implicated in nociceptor hyperactivity, and many protein kinases in addition to PKA can activate CREB), and each has additional interactions with the same and other cell signaling pathways. Only steps for which biochemical and/or pharmacological evidence is available are indicated. Abbreviations: 5-HT, serotonin; 5' UTR, 5' untranslated region; CREB, cAMP response element-binding protein; eIF4E, eukaryotic translation initiation factor 4E (correspondingly for eIF4G, eIF4A); ERK, extracellular signal-regulated kinase; GPCR, G protein-coupled receptor; G_s, stimulatory G protein; MEK, mitogen-activated protein kinase kinase; MNK, mitogen-activated protein kinase-interacting kinase; P, phosphorylation; PABP, poly-A-binding protein; PKA, protein kinase A; Raf, rapidly activated fibrosarcoma kinase; Ras, rat sarcoma GTPase. Created with [BioRender.com](https://www.biorender.com).

depolarization and consequent C-Raf and ERK signaling, plus a reduction of adenylyl cyclase's sensitivity to inhibition by opioids [115]. In addition, CREB activity is important for persistent nociceptor sensitization in rodents [116,117]. Thus, MNK-dependent protein synthesis, ERK signaling upstream of MNK [118], cAMP signaling, and CREB activity are associated with nociceptor hyperactivity in species (Table 1) whose last common ancestor lived ~600 million years ago.

Human nociceptors may be especially prone to hyperactivity

Sometime during the past 65 million years, after primates and rodents diverged from the common ancestor of placental mammals [119], notable modifications in nociceptors evolved, as reflected in several key differences between nociceptors in humans and rodents. Rodent nociceptors are

typically divided into two distinct classes, peptidergic and non-peptidergic. Peptidergic nociceptors express TrkA receptors into adulthood, as well as neuropeptides such as calcitonin gene-related peptide (CGRP); they typically innervate deeper tissues and connect to neurons in lamina I of the spinal dorsal horn. Non-peptidergic nociceptors innervate the skin, do not express TrkA in adulthood, and connect to neurons in lamina II of the dorsal horn. The two classes have specializations for thermal and mechanical pain, respectively [120], although genetic tools are revealing finer functional subdivisions within each class. In contrast, all adult human nociceptors express TrkA and most express CGRP, suggesting that human nociceptors are generally peptidergic [121,122]. Unlike rodent nociceptors, among which ~50% express TRPV1, all human nociceptors express TRPV1. TRPV1 is readily sensitized by diverse inflammatory mediators such that its temperature activation threshold falls into the range of core body temperature [123], producing ongoing AP discharge (Figures 1D and 2B). In addition, a larger fraction of probable nociceptors in humans expresses one or more acid-sensing ion channels (ASIC1, 2, and 3) [124]. Thus, human nociceptors appear more 'polymodal' than rodent nociceptors, and possibly than other primate nociceptors [122,125–127]. This should make them sensitive to a broader range of sensory stimuli and inflammatory mediators, thereby increasing extrinsically driven nociceptor hyperactivity and more readily inducing intrinsic hyperactivity (Figure 2).

Additional observations suggest that human nociceptors may be more intrinsically excitable than rodent nociceptors. First, dissociated human DRG neurons exhibit a greater density of tetrodotoxin (TTX)-sensitive and TTX-resistant Na_v current than rodent nociceptors [128]. Second, RNA sequencing and functional experiments suggest that expression of the TTX-sensitive channel Nav1.7 is higher in human than rodent nociceptors [129]. Third, TTX-sensitive current is activated at more hyperpolarized potentials in human nociceptors [128]. Fourth, there is little or no use-dependent inactivation of TTX-resistant currents in human nociceptors [128]. All these differences should further promote spontaneous activity in human nociceptors, as observed *in vivo* [2,7] and in DRG neurons dissociated from neuropathic pain patients [84,104]. Interestingly, different densities and kinetics of human voltage-gated calcium (Ca_v) currents suggest that activity-dependent increases in intracellular Ca^{2+} may be smaller in human than rodent nociceptors [130]. If human evolution led to a high propensity for persistent nociceptor hyperactivity, then smaller Ca^{2+} responses might help protect against toxicity from excessive Ca^{2+} accumulation during continuing hyperactivity.

Could a high susceptibility to nociceptor-driven persistent pain in humans be evolutionarily adaptive? Although other species can display apparent empathy and helping behavior [131–133], humans are unique in the extent and flexibility of their helping behavior [134,135]. Thus, even early humans may have provided more help to injured family and group members than occurs in most species. Furthermore, humans are much more likely to communicate their pain to group members than are other species [136,137]. An intriguing possibility is that persistent nociceptor hyperactivity and other traits that maintain painful awareness of injury-induced vulnerability were subject to increased natural selection in hominins. The evolution of more flexible and complex hominin social behaviors may have enabled more effective and longer-lasting aid and protection for those needing help, with help-seeking being spurred by nociceptor-driven pain in humans impaired by injury (see [136]).

Concluding remarks

Studies in both mammals and molluscs (Table 1) show that injuries sufficiently severe to damage nerves can induce persistent nociceptor hyperactivity. Findings in squid and mice suggest that persistently enhanced vulnerability resulting from impairment after injury can be lessened by behavioral alterations, including hypervigilance driven by prolonged nociceptor hyperactivity.

Outstanding questions

What are the relevant biological needs shared by animals exhibiting persistent nociceptor hyperactivity? Do species that exhibit little inflammatory or neuropathic pain, such as naked mole-rats, lack persistent nociceptor hyperactivity after significant injuries?

To what extent are biophysical and molecular mechanisms (intrinsic and extrinsic, including persistent immune alterations) that drive persistent nociceptor hyperactivity conserved across vertebrate and invertebrate species, and among mammalian species?

Do types of primary afferent neurons other than nociceptors (e.g., pruriceptors and low-threshold mechanosensory neurons) exhibit persistent hyperactivity that might be adaptive in some conditions?

Are the apparently high excitability and high sensitivity to extrinsic pain-related signals in human nociceptors linked to a high susceptibility of humans to persistent nociceptor hyperactivity and consequent pain?

Many clinical conditions in humans cause neuropathy, altered sensory function, and chronic pain (e.g., diabetes, chemotherapy, radiculopathy, postherpetic neuralgia). To what extent does pain in such conditions involve accidental recruitment of a persistent nociceptor hyperactivity state that evolved originally as an adaptation to severe peripheral injury versus purely pathological effects of neuropathy, such as metabolic dysfunction?

The lack of clear evidence thus far for injury-induced persistent nociceptor hyperactivity in naked mole-rats, *Drosophila*, and *C. elegans* suggests either that a protective hypervigilance function has not been evolutionarily significant in some lineages (e.g., because of limited predation or short lifespans) or that alternative adaptations (e.g., central sensitization) were preferentially selected for this function (Table 1). In rodent and human nociceptors, synergistic electrophysiological alterations underpinning spontaneous activity are consistent with hyperactivity functioning to drive persistent hypervigilance. Current findings emphasize the need for further comparisons of species having and lacking persistent nociceptor hyperactivity (see Outstanding questions). Also needed is expanded investigation across taxa into common as well as divergent cellular and molecular mechanisms in neurons, glia, and immune cells driving persistent hyperactivity in nociceptors (and possibly in other types of primary somatosensory neurons). Finally, differences in sensitivity and excitability between nociceptors in humans and other mammals encourage further study into the possibility that, during recent hominin evolution, nociceptors increased their susceptibility to persistent hyperactivity, increasing the propensity of humans to chronic pain.

Acknowledgments

E.T.W. was supported by National Institutes of Health (NIH) grants NS091759 and RO1NS111521. R.J.C. was supported by an Allen Distinguished Investigator Award, a Paul G. Allen Frontiers Group advised grant of the Paul G. Allen Family Foundation, and NSF IOS 204-7331. G.G.N. was supported by Australian National Health and Medical Research Council grants APP1026310, APP1028887, and APP1046090. T.J.P. was supported by NIH grants NS111929 and NS102161. E.S.S. was supported by a grant from the Medical Research Council MR/W002426/1.

Declaration of interests

T.J.P. is a co-founder of 4E Therapeutics, Doloromics, PARMedics, and NuvoNuro; he serves on the Board of Directors of 4E Therapeutics and Doloromics; he is an inventor on patents related to MNK inhibition for the treatment of pain. E.S.S. receives funding from AstraZeneca and GlaxoSmithKline. E.T.W., R.J.C., and G.G.N. declare no competing interests.

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